

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8504**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600053
Course Section:	1				

L 29, 510, 986, 1, 1505, 38441



Name of the candidate. Asim Homal Shah

Candidate's UID number:

2	0	2	2	6	0	0	0	5	3
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Examination class room number:

4	0	1
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Invigilator's signature with date:

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(To be filled in by the candidate)
EXAMINATION: BE - BTECH MSE
(For example FY MCA/FY BTECH)
PROGRAM: CSE (AIML)
DAY AND DATE: 8th October 2025
Wednesday
COURSE NAME: Deep Learning

SEMESTER : 7
I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



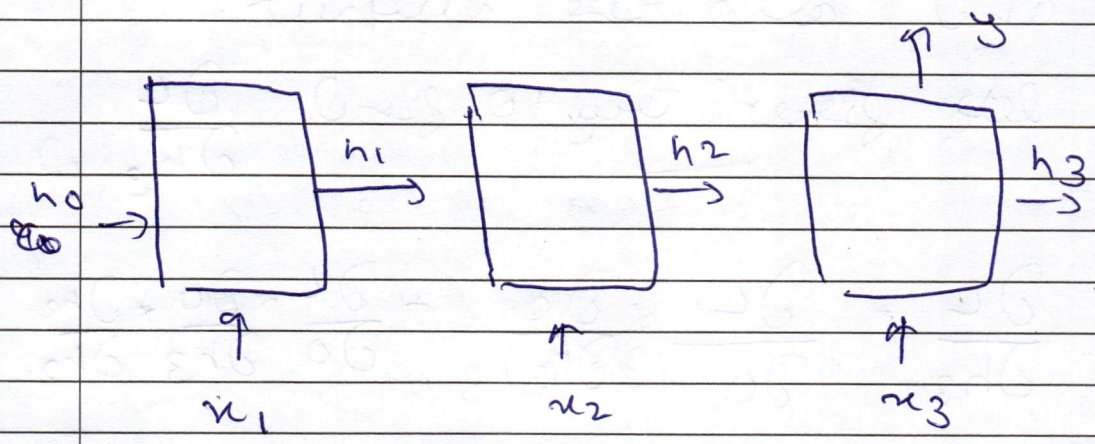
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Question
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Q3 a

first let's draw the structure
of recurrent neural network



we know that for recurrent
neural networks the hidden
state is calculated as
follows

$$h_t = \tanh(w_{hh} \times h_{t-1} + w_{nh} \times x_t + b_h)$$

and the output is calculated
as

$$y = \text{Softmax}(w_{hy} \times h_t + b_y)$$

as we have seen that when
we try to find the ~~derivative~~
derivative with respect to

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the hidden state

$$\frac{\partial L}{\partial h_{t-k}}$$

the hidden state depends on its subsequent inputs from the previous timestamps and also the output

lets first try to find $\frac{\partial L}{\partial h_3}$

$$\frac{\partial L}{\partial h_3} = \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial h_3} + \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial h_3} \times \frac{\partial h_3}{\partial h_2}$$

$$+ \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial h_3} \times \frac{\partial h_3}{\partial h_2} \times \frac{\partial h_2}{\partial h_1}$$

this follows a pattern which we can write as

$$\frac{\partial L}{\partial h_3} = \sum_{i=1}^3 \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial h_i}$$

now to generalize this

$$\frac{\partial L}{\partial h_{t-k}} = \sum_{i=1}^{t-k} \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial h_i}$$



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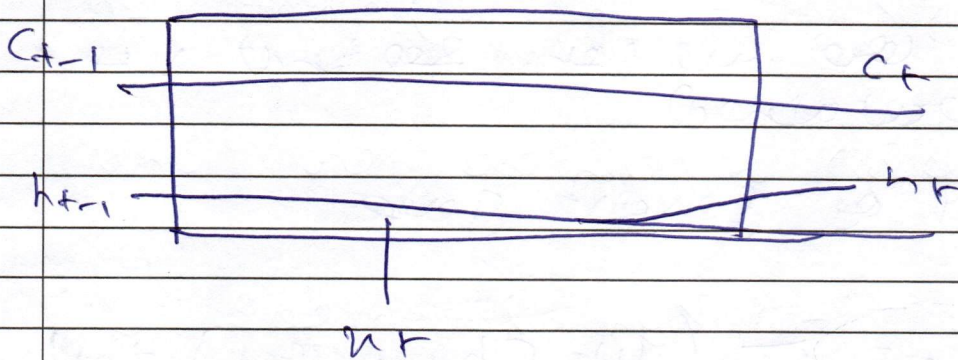
Here we see that all the gradients from the previous timestamps are multiplied continuously

for eg

$$\frac{\partial L}{\partial \theta} \times \frac{\partial \theta}{\partial h_3} \times \frac{\partial h_3}{\partial h_2} \times \frac{\partial h_2}{\partial h_1}$$

If the time stamp is large the value of the gradient will be close to 0 which will cause vanishing gradients.

To prevent this we have LSTM. Here we have 2 memories long term and short term which allows us to capture some long term dependencies thus providing us with a pathway for terms in previous timestamps to stay.



Equation of C_t

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t$$

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now & here comes forget gate of this allows the values from previous time steps to have a say this is because we forget ~~gate~~ gate used sigmoid activation function which has its values ranging from 0 to 1

due to this we can forget some things and retain others by keeping the value close to 0 and 1 respectively

eg

$$\begin{bmatrix} 0.9 \\ 0.1 \\ 0.8 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.7 \\ 0.2 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 g_t ch_{t-1} i_t o_t

here we can see that 5 is retained

eg of forget gate

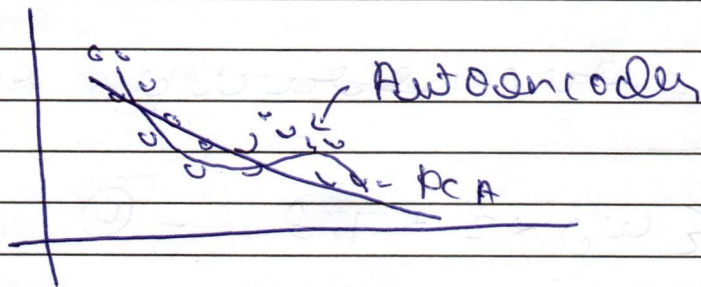
$$g_t = \sigma(w_g[ch_{t-1}, x_t] + b_g)$$

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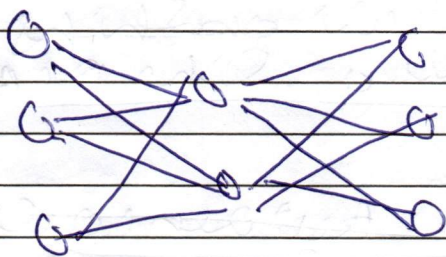
Q2 b

an Autoencoder unlike PCA can be used to capture non linear relationships due to the nature of its non linearity



However if we use a linear activation function the model's capacity to capture non linear equations is reduced. Due to which it will behave like PCA.

Let's take a simple example



Here the equation if we use linear functions will be like:-

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1] encoder part

 ~~$w \cdot x$~~

~~$$z = w \cdot x + b$$~~

1] latent space (z)

$$z = w \cdot x + b \quad - (1)$$

2] reconstruction part

$$x = w \cdot z + b \quad - (2)$$

Sub ① in 2

equation of autoencoder
with linear function

~~$$x = w \cdot (w \cdot x + b) + b$$~~

as we can see it's a
linear function thus
not able to capture
complex relationships
and learning like PCA

Over complete Autoencoder
is a autoencoder which
has the dimensionality
of its bottleneck greater
than the input

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	6	
0	0	6
0	0	0
0	0	0
n	0	5
	2	

Here now the problem
is that since ~~since~~

$\dim(n) > \dim(z)$ ~~instead~~.
Instead of learning twice,
features of n the model
can ~~just~~ ~~create~~ create
by copying n as it is to z
eg $n = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

	6	
①	①	60
②	②	0
③	③	0
	③	

(Double copied)

Thus we need a ~~regular~~
regularization ~~term~~ ~~term~~
term which will penalize
this & hence all our
prevent creating by the
model



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Q7a

Q7b

we know that bias is the error caused by the model making strong assumptions about the data

Variance is the error caused when a ~~strong~~ ~~variety~~ slight variance in training data can lead to a huge error

The decomposition

$$\text{error} = \text{bias}^2 + \text{variance} + \text{noise}$$

now if the model's capacity is too high it will train on the ~~too~~ data and the ~~high~~ point where it starts memorizing the data instead of learning. Due to this it learns the outliers as well

now since the model is strongly dependent on the training data even a slight ~~change~~ change will lead to a error

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There is an increase in the
version of the model
which ~~increases~~ ~~the~~ ~~version~~
increases the version



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Q3b

GRU has mainly two types of gates

1] update gate:- which all over for retention of information

2] reset gate :- which all over to forget information

Here the equation of update gate is

$$u_t = \sigma(W_u[n_{t-1}, n_t] + b_u)$$

This is because u_t will depend on the previous state and current input to determine what we want to retain

Similarly r_t depends again, depend on the previous state and current input to determine what to forget so the equation of reset gate is

$$r_t = \sigma(W_r[n_{t-1}, n_t] + b_r)$$

Here σ is used to keep values between 0 and 1

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~~that final state~~

now if we compare LSTM
and GRU we see that

to compute the output
of GRU we need
4 calculations in ~~the~~ ^{each} ~~cell~~ ^{unit}
if calculate

forget gate, update gate,
candidate value and hidden
state

To calculate the output of LSTM
we need 6 calculations

input gate, forget gate, output gate,
candidate value, long term
memory and hidden state

thus we can see that GRU
is more computationally
feasible since it has
reduced calculations
which means it has reduced
for smaller tasks we can
train this for a
long time improved generalization
in real world
applications like, natural
language processing, machine
translation, sentiment analysis, name
generation etc

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Q79

$$f(w) = \frac{1}{2} (xw - y)^2$$

$$f'(w) = \frac{1}{2} 2 (xw - y) = (xw - y)$$

answer on simplifying
~~simplifying~~ ~~gives~~

$$f'(w) = (xw - y) (1 - yw)$$

thus the update rule
is

$$w_{new} = w_{old} - \eta \cdot f'(w)$$

$$w_{new} = w_{old} - \eta \cdot (xw - y) (1 - yw)$$

now if $n > \frac{2}{\epsilon_{max}}$ then

~~The gradient will
decrease because
its eigen value is
very large~~



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The gradient will overshoot
due to it being large

thus the correction factor
must be less than $\frac{2}{\lambda_{max}}$

Q22

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Q24

In standard 2D convolution the computational complexity is squared because each filter will take ± 2 time to do one calculation and to do it to the entire image it will be $\pm 2 \times H \times W$ this is for 1 filter

now if we have k of them then time complexity is $O(\pm 2 \times H \times W \times k)$ which is $O(\pm 2)$.

Depth wise convolution overcome this by having linear time complexity $O(k)$. Due to this they are very useful in modern architectures. This is because modern applications which require speed and fast results like a real time image

classification model. A Depth wise convolution network will provide speed which will lead to faster, reliable they are being used in modern applications.

Also due to the efficiency of depth wise convolution network it has less parameters. This is more efficient as compared to vanilla convolutional networks.